



Noise Waves

$$\vec{b} = \bar{S} \vec{a} + \vec{c}$$

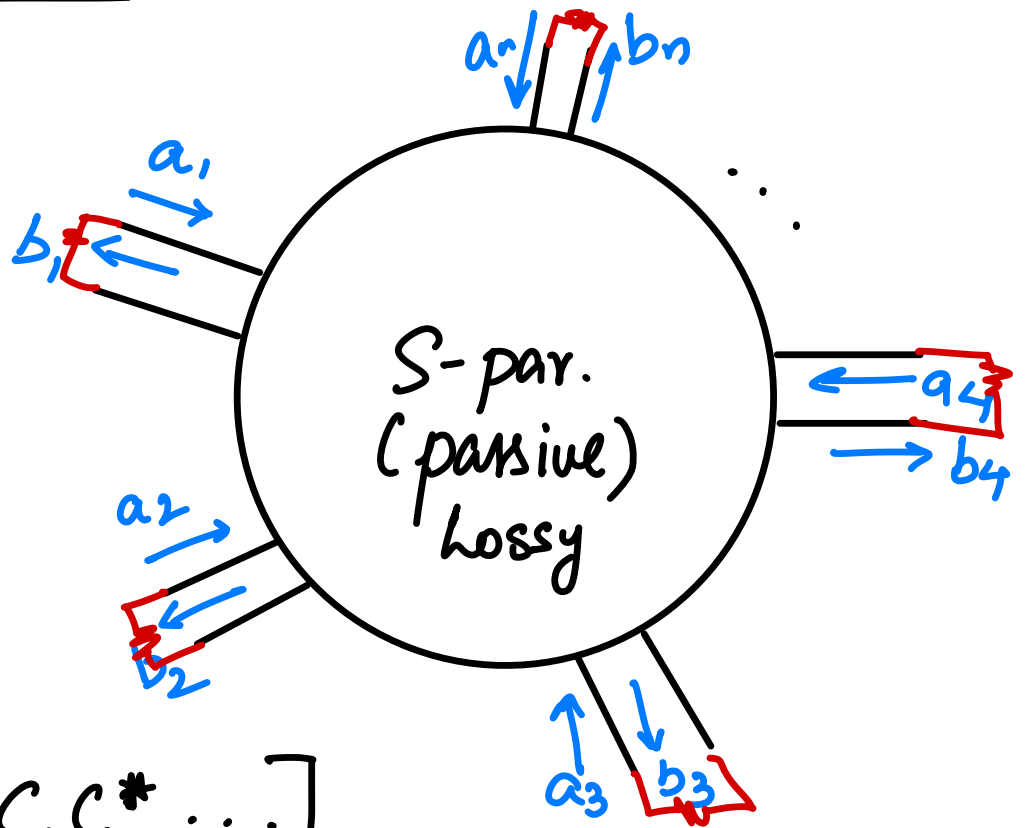
$$C_s = \overline{\vec{c} \vec{c}^+}$$

$$= E[\vec{c} \vec{c}^+]$$

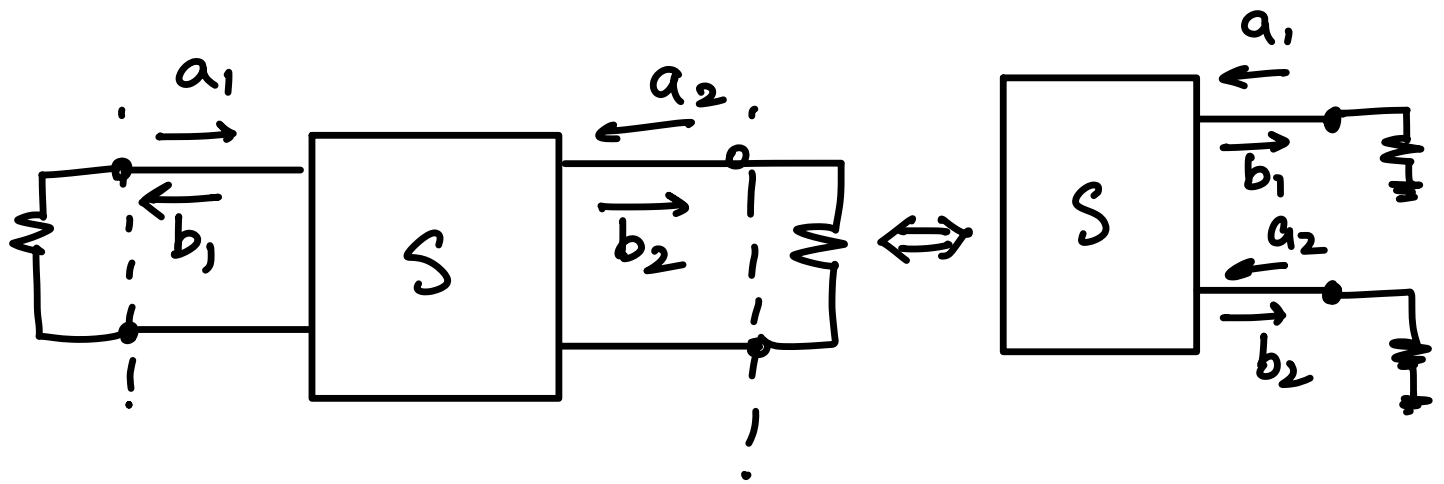
$$= E \begin{bmatrix} C_1 C_1^* & C_1 C_2^* & \dots \\ C_2 C_1^* & \ddots & \end{bmatrix}$$

$$\Rightarrow C_s = \begin{bmatrix} \overline{C_1 C_1^*} & \overline{C_1 C_2^*} & \dots \\ \overline{C_2 C_1^*} & \ddots & \end{bmatrix}$$

Noise Correlation Matrix.



Passive Two Port



Assumptions

- ① Thermal equilibrium.
- ② Reflectionless terminations.

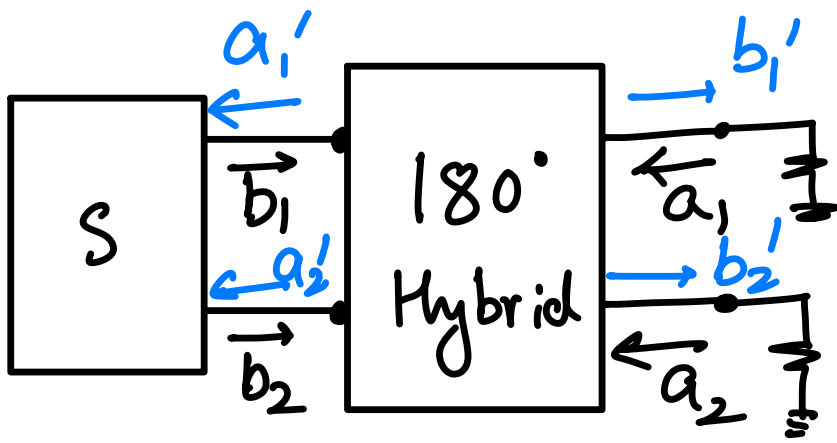
$$\Rightarrow \overline{|a_1|^2} = \overline{|a_2|^2} = KTB$$

$$\times \overline{a_1 a_2^*} = 0$$

$$\Rightarrow \overline{|b_1|^2} = \overline{|b_2|^2} = KTB$$

but what about $\overline{b_1 b_2^*} = ?$

We will show that $\overline{b_1 b_2^*} = 0$.



$$b_1' = \frac{1}{\sqrt{2}} (b_1 + b_2) \quad b_2' = \frac{1}{\sqrt{2}} (b_1 - b_2)$$

$$\overline{|b_1'|^2} = \overline{\left[\frac{1}{\sqrt{2}} (b_1 + b_2) \right] \left[\frac{1}{\sqrt{2}} (b_1^* + b_2^*) \right]}$$

$$= \frac{1}{2} \left[\overline{b_1 b_1^*} + \overline{b_2 b_1^*} + \overline{b_2 b_2^*} + \overline{b_1 b_2^*} \right]$$

$$= \frac{1}{2} \left[\overline{|b_1|^2} + \overline{|b_2|^2} \right] + \text{Re} \left[\overline{b_2 b_1^*} \right]$$

$$\overline{|b_2'|^2} = \frac{1}{2} \left[\overline{|b_1|^2} + \overline{|b_2|^2} \right] - \text{Re} \left[\overline{b_2 b_1^*} \right]$$

$$\overline{b_1' b_2'^*} = \frac{1}{2} \left[\overline{|b_1|^2} - \overline{|b_2|^2} \right] + j \text{Im} (\overline{b_2 b_1^*})$$

Similar expressions for

$$\overline{|a_1'|^2}, \overline{|a_2'|^2}, \overline{|a_1' a_2'^*|}$$

$$\Rightarrow \overline{|a_1'|^2} = \overline{|a_2'|^2} = kTB$$

$$\times \overline{|a_1' a_2'|} = 0$$

$$\text{Thermal Eq.} \Rightarrow \overline{|b_1|^2} = \overline{|b_2|^2} = kTB.$$

$$\Rightarrow \overline{|b_1'|^2} = kTB + \text{Re} [\overline{b_2 b_1'^*}]$$

$$\overline{|b_2'|^2} = kTB - \text{Re} [\overline{b_2 b_1'^*}]$$

$$\text{Thermal Eq.} \Rightarrow \text{Re} [\overline{b_2 b_1'^*}] = 0$$

$$\text{Using a } 90^\circ \text{ hybrid} \Rightarrow \text{Im} [\overline{b_2 b_1'^*}] = 0$$

$$\Rightarrow \overline{b_2 b_1'^*} = 0$$

Bosma's Theorem

$$\overline{\vec{a} \vec{a}^+} = KTB \overline{\overline{\vec{I}}}$$

and $\overline{\vec{b} \vec{b}^+} = KTB \overline{\overline{\vec{I}}}$ Since $\overline{b_i b_j^+} = 0$
for $i \neq j$

$$\vec{b} = \overline{\vec{S}} \vec{a} + \vec{c}$$

$$\begin{aligned} \Rightarrow \underbrace{\overline{\vec{b} \vec{b}^+}}_{KTB \overline{\overline{\vec{I}}}} &= \overline{(\overline{\vec{S}} \vec{a} + \vec{c})(\overline{\vec{S}} \vec{a} + \vec{c})^+} \\ &= \overline{(\overline{\vec{S}} \vec{a} + \vec{c})(\vec{a}^+ \overline{\vec{S}}^+ + \vec{c}^+)} \\ &= \overline{\overline{\vec{S}} \underbrace{\vec{a} \vec{a}^+}_{KTB \overline{\overline{\vec{I}}}} \overline{\vec{S}}^+ + \underbrace{\vec{c} \vec{c}^+}_{C_s}} \end{aligned}$$

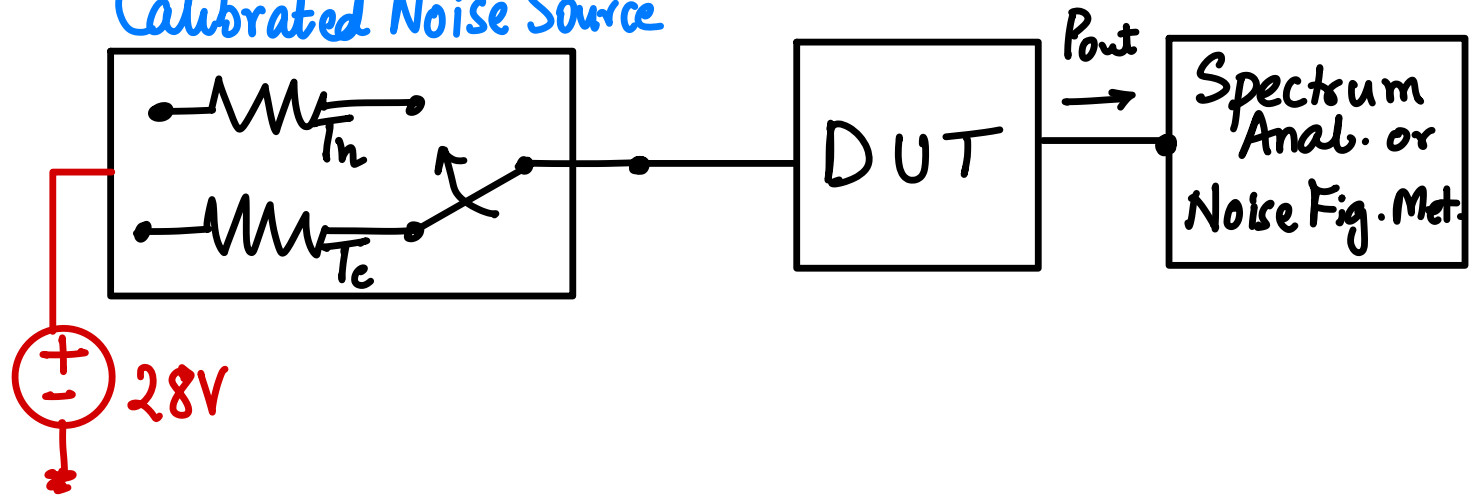
$$\Rightarrow \boxed{C_s = KTB (\overline{\overline{\vec{I}}} - \overline{\vec{S}} \overline{\vec{S}}^+)}$$

BOSMA'S THEOREM.

Noise Figure Measurements

Y-factor Method

Calibrated Noise Source



Excess Noise Ratio

$$ENR = \frac{T_h - T_c}{T_c} \quad (\text{is a fn. of freq})$$

Y-factor Method

$$Y = \frac{P_{out}^h}{P_{out}^c}$$

$$P_{out}^c = G (k T_0 B) + N_a$$

$$\Rightarrow F = \frac{ENR}{\gamma - 1} \quad \text{if } T_c = T_0$$

Proof:

$$\gamma = \frac{P_{out}^h}{P_{out}^c} = \frac{KT_hBG + Na}{KT_cBG + Na}$$

$$= \frac{\frac{T_h}{T_0} + \left(\frac{Na}{KT_0BG} \right) \rightarrow F-1}{\frac{T_c}{T_0} + \frac{Na}{KT_0BG}}$$

$$\Rightarrow F = \frac{T_h - \gamma T_c + T_0(\gamma - 1)}{T_0(\gamma - 1)}$$

$$\text{If } T_c = T_0$$

$$F = \frac{T_h - T_c}{T_0} \cdot \frac{1}{\gamma - 1}$$

$$\boxed{F = \frac{ENR}{\gamma - 1}}$$

Measurement Procedure

Step 1 Calibration

> Connect noise source directly to the receiver & get Y_r

$$\Rightarrow F_r = \frac{ENR}{Y_r - 1}$$

Step 2: DUT + Receiver

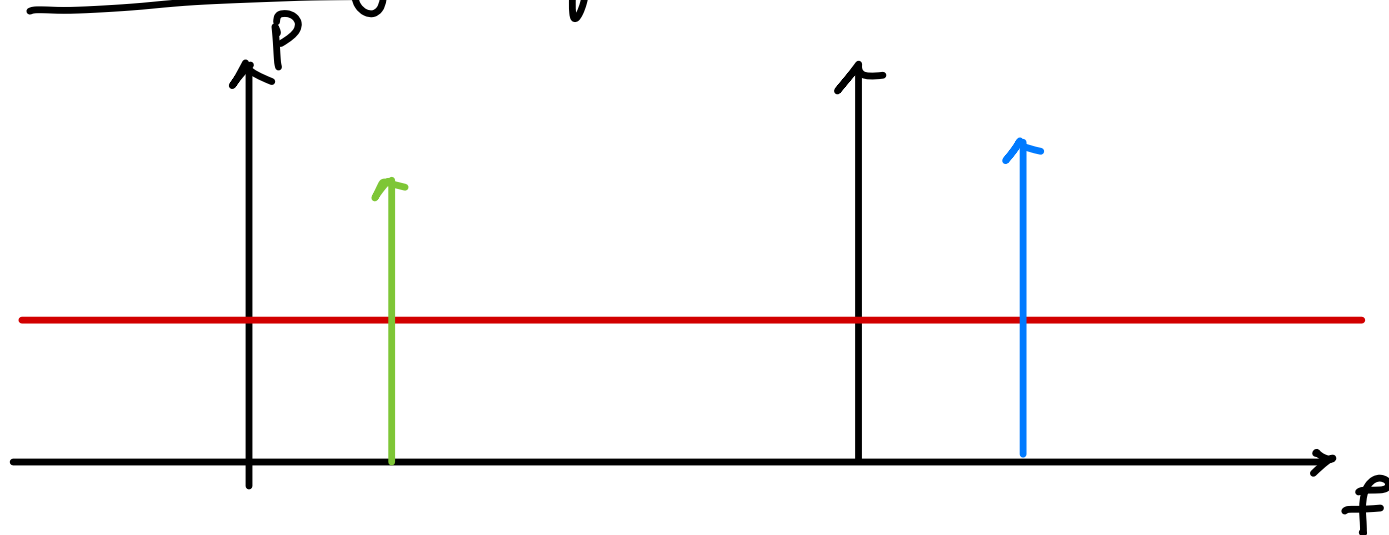
$$F_{sys} = \frac{ENR}{Y_{sys} - 1}$$

$$\Rightarrow F_{DUT} = F_{sys} - \frac{F_r - 1}{G_{DUT}}$$

$$G_{DUT} = \frac{F_{sys}}{F_r} \cdot \frac{P_{out,sys}^c}{P_{out,r}^c}$$

Limitation: Not very accurate when $NF \uparrow$ or $G \downarrow$.

Noise Figure of Mixers



SSB Noise Figure

Assumes signal is in one sideband & noise is in both sidebands.

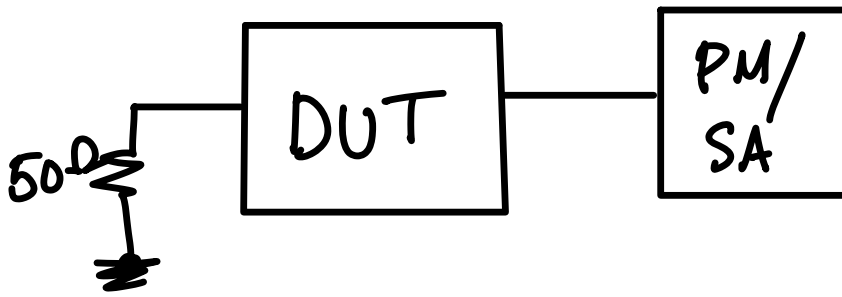
DSB Noise Figure

Assumes signal & noise are in both sidebands or in only one sideband.

CAUTION: Y-factor method always gives the DSB Noise Figure.

$DSB_{NF} < SSB_{NF}$ by 3dB.

Gain Method



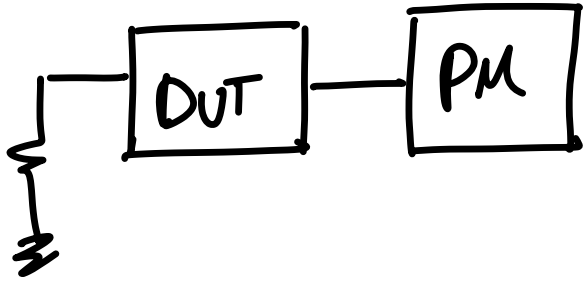
$$P_{out,sys} = G K T_0 B + N_a$$
$$= F (G K T_0 B)$$

$$\Rightarrow F_{sys} = \frac{P_{out,sys}}{G K T_0 B}$$

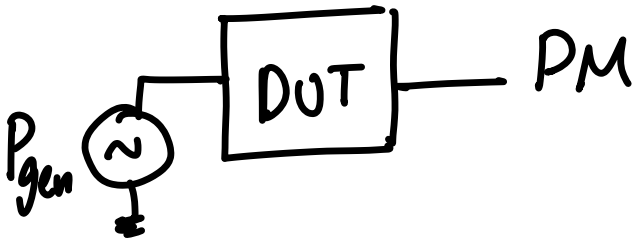
$$NF = 10 \log (P_{out,sys}) - 10 \log (G)$$
$$- 10 \log (K T_0 B)$$

Use Noise Marker to get P_{out} !

Signal gen twice power method



$$P_{out}^{(1)} = G K T_0 B + N_a = F (G K T_0 B)$$



$$P_{out}^{(2)} = G K T_0 B + N_a + G P_{gen}$$

$$\Rightarrow P_{out}^{(2)} = 2 P_{out}^{(1)}$$

$$\Rightarrow G K T_0 B + N_a + G P_{gen} = 2 (G K T_0 B + N_a)$$

$$\Rightarrow G P_{gen} = F G K T_0 B$$

$$\Rightarrow \boxed{F_{sys} = \frac{P_{gen}}{K T_0 B}}$$